



Eco-Friendly Materials for the Removal of Some Heavy Metals from Contaminated Water

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Abstract: Industrialization and population growth pose significant environmental issues, particularly in water quality, making many sources unsuitable for domestic use. Natural organic compounds and metal nanoparticles (NPs) are used as wastewater adsorbents. The current research investigated the adsorption kinetics, isotherms, and reusability of the manganese oxide NPs synthesized from star anise (SA) (*Illicium verum*) extract (MnO@SE) to aid in the creation of environmentally friendly water purification solutions. MnO@SE was prepared with SA extract and manganese acetate (II) tetrahydrate solution. The green-synthesized biosorbent was characterized employing methods including Fourier transform infrared spectroscopy, X-ray diffraction (XRD), energy-dispersive X-ray spectroscopy (EDX), and scanning electron microscopy (SEM). These evaluations offered good information on surface shape and surface-available functional groups. The influences of pH, adsorbent dosage, ion concentration, and contact time on metal ion adsorption were all examined. The results revealed that model solutions with a pH of 2.0, a biosorbent dosage of 0.8 g/L, an initial concentration of 25 mg/L, and a contact time of 50 minutes produced the best removal efficiency (96.34% for Cr(VI) and 87.01% for Pb(II)). The adsorption processes of both metal ions occurred in a multilayer fashion on the heterogeneous surface of the biosorbent through diffusion kinetics, according to the isotherm and kinetic findings. The adsorption process is endothermic and spontaneous, according to thermodynamic analysis. The study revealed that the green-synthesized MnO@SE effectively removed 96.34% Cr(VI) and 87.01% Pb(II) under optimal conditions, promoting eco-friendly water purification through multilayer, endothermic, spontaneous, and diffusion-driven adsorption.

Keywords: Environmentally friendly; Nano-composite; Green synthesis; Heavy metals; Absorption; Star anise; Kinetics studies

1 Introduction

Escalating industrial development combined with demographic growth contributes substantially to environmental problems, particularly in water quality due to unregulated by-product release, making many sources unsuitable for domestic use [1]. Polluted industrial wastewater and industrial wastewater discharged from untreated power plants also pose a prominent environmental problem attributed to the rapid proliferation of industries, which leads to increased accumulation of harmful pollutants in water resources [2]. Moreover, untreated sewage and incompletely treated industrial wastewater from factories, plants, and power plants, which seeps into groundwater, speeds up the decline in the standard of water resource quality. To improve the environment, the effects of these pollutants must be addressed and reduced. Industrially used heavy metals, dyes, and inorganic solvents contaminate water and, at low concentrations, also affect aquatic organisms and human health [2, 3]. Heavy metals from non-biodegradable substances like tanning, electroplating, paper, and fertilizer manufacturing significantly increase water toxicity, presenting substantial risks to

ecological systems as well as human welfare [2, 4]. Global freshwater shortage necessitates new technologies for effective treatment, as traditional methods struggle to eliminate heavy metal pollution [2, 5, 6].

Current treatment methods for inorganic water contaminants include chemical precipitation, ion exchange, adsorption, electrocoagulation, and membrane separation but they have disadvantages like high energy usage and incomplete purification. Adsorption serves as a superior approach for the removal of non-biodegradable impurities from wastewater, as it adheres to an adsorbent surface via van der Waals forces, reducing secondary sludge production [7]. Also lead (Pb) and cadmium (Cd) are two of the most common pollutants that exert an adverse impact on the quality of soil and water. Even though they are widely used in many different industries, such as mining, manufacturing, and inappropriate waste disposal techniques, all release them into the environment. These metals are dangerous owing to their capacity to accumulate along the food chain and cause acute poisoning or long-term exposure that can harm living things, including humans [2, 8].

Nanoscience and nanotechnology are crucial in modern material science, focusing on nanoparticles (NPs; 1–100 nm size) with unique electrical, magnetic, and catalytic properties and high surface area [9]. Metal oxide NPs are widely used in various technologies. Particularly, manganese oxide (MnO) NPs have gained significant attention. Culture-based isolation and polymerase chain reaction (PCR)-based molecular identification methods were used to identify heavy metal-resistant bacteria. These NPs and methods are applied in magnetic data storage, catalysis, electrodes, molecule adsorption, sensors, and electronics [8, 10]. Manganese oxide nanostructures, created through various methods, offer low toxicity, cost-effectiveness, and eco-friendliness for optimizing polymer electrolytes' ionic conductivity and mechanical and thermal characteristics [11]. Therefore, the co-precipitation method is the most commonly applied method due to its simplicity, rapid preparation, cost-effectiveness, particle size control and surface modification [12].

Natural organic compounds, like fruit peel, bark, roots, seeds, and plant extracts are commonly used as wastewater adsorbents, with significant research on metal NPs [13]. With an extremely high surface-to-volume ratio, they are extremely antibacterial active and effective for the adsorption of heavy metals, like Pb^{2+} , Cd^{2+} , Cu^{2+} , and Ni^{2+} . Manganese oxide is prepared by chemical, physical, and biological methods. Chemical reduction is cost-effective and straightforward but poses environmental risks due to toxic reagents, while plant bioreductant are less toxic and environmentally friendly [14]. Plant secondary metabolites, like saponins, terpenoids, flavonoids, and tannins, are employed for the preparation of MnO NPs [15]. It has been successfully employed to prepare manganese NPs from diverse plant-derived extracts, like *Artemisia annua*, *Perilla frutescens*, and plantain peels [16]. The present study used star anise (SA) to synthesize MnO NPs, a green pathway, for the elimination of chromium (Cr) and Pb ions through batch adsorption. The study explores the mechanisms of adsorbing synthetic heavy metal solutions using this green adsorbent. For determining the physical and chemical properties of MnO NPs prepared by green synthesis, specific analytical equipment was employed, including Fourier transform infrared spectroscopy (FTIR), scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDX), and X-ray diffraction (XRD) analysis. Adsorbent dose, initial concentration, reaction time, and pH were studied under different conditions to determine the optimum adsorption efficiency.

The current research aimed to explore the adsorption kinetics, isotherms, and reusability of the synthesized MnO nanocomposites, contributing to the creation of environmentally friendly approaches for water treatment.

2 Materials and Methods

2.1 Source of Plant Material

Dry SA fruit was obtained from a local food retail vendor from Al-Sadriya market. The plant material was identified in the Department of Biology, Ibn Al-Haytham College of Science, University of Baghdad, and authenticated by Assistant Professor Dr. Arej Abdalsatar Farman, a taxonomist at the Herbarium Unit (Voucher No.: ILV-2025-001).

2.2 Preparation of Plant Extract

To prepare the plant extract, 50 mL of deionized water was introduced to 2 g of powdered SA in a flask, and the mixture was heated for 2 minutes. After stirring the solution and allowing it to cool, the clear supernatant, designated as the SA extract (SE), was discarded using a centrifuge set to 7,500 rpm for 10 minutes. The solution was then kept at 4 °C until use.

2.3 Green Synthesis of Manganese Oxide Nanoparticles

The synthesis of MnO NPs was achieved by boiling powdered SA (2 g) with deionized water (50 mL). The mixture was stirred for 2 minutes and then centrifuged at 7,500 rpm for about 10 minutes. The clear supernatant was discarded, and the remaining supernatant was stored at 4 °C. A modified method from Shanshool and Shanan [17] involved combining a 40 mM solution of manganese acetate (II) tetrahydrate with 1 mL of SE. The mixture was incubated at 25 °C. The study aimed to synthesize MnO NPs by varying physicochemical parameters, such as SE concentrations, metal ion volumes and concentrations and incubation temperature. Metal ion levels were systematically altered

from 1 mM up to 80 mM, and 1 mL of each concentration was incubated with 1 mL of SE. The SE volume was progressively varied between 0.25 mL and 1 mL. Furthermore, the influence of the volume of the metal ion on the yield and particle size of MnO NPs was evaluated by varying metal ion volumes ranging between 1 mL and 5 mL and varying SE volume from 1 to 4 mL at a fixed volume of metal ions of 1 mL. Influences of incubation temperature on the properties of NPs were determined by subjecting a mixture of 1 mL of 40 mM metal ion solution and 1 mL of SE to incubation conditions at temperatures ranging from 25 °C to 85 °C.

2.4 Batch Adsorption Study

Adsorption performance of the MnO NPs synthesized from SA extract (MnO@SE) was investigated by batch adsorption experiments at various operating conditions. A 100 mL simulated solution containing simulated Cr(VI) and Pb(II) ions was made. Solution pH was regulated as required by 0.1 M sodium hydroxide (NaOH) or 0.1 M nitric acid (HNO₃), and 0.1 g of MnO@SE adsorbent was used in the solution for trial experiments. The solution was shaken vigorously on a shaker at 150 rpm for one hour to allow proper interaction between adsorbent and metal ions and the pH value of the final solution was measured after adsorption time at room temperature (25 °C) with a pH electrode. The adsorbent was separated from the treated solution employing cellulose acetate filter paper with 0.45 μm pore size. Atomic absorption spectroscopy (AAS) was used to determine the initial concentration (C_i) and equilibrium concentration (C_e) of metal ions in the simulated solution. The adsorption capacity of MnO@SE was determined by adjusting pH with HNO₃, and shaken vigorously. An adsorbent (0.01 g) was used in combination with 0.01 mol/L solutions of NaOH, and 2 hours of nitrogen gas were passed through the combined solution.

The pH of a deoxygenated sample was adjusted from 2.0 to 7.0 with ongoing agitation for two days, and initial and final pH were recorded and plotted to find the zero point of charge (pH_{ao}), i.e., where initial and final pH are equal. Subsequently, the impacts of contact time on adsorption performance were tested with 0.1 g of MnO@SE adsorbent in the simulated solution, which was set at the optimum pH. Contact times ranging from 10 minutes to 2 hours under room temperature were utilized to analyze variations in adsorption performance with time. Adsorption performance was also investigated further by changing the adsorbent dosage from 0.1 g to 0.5 g while maintaining the optimal pH and contact time obtained in the previous experiment's constant. The experiment was done at 25 °C with a constant initial concentration of the metal ion being maintained at 50 mg/L. Lastly, the influence of changing initial metal ion concentration from 25 to 100 mg/L on the impact of adsorption was studied at 25 °C using the already optimized pH, MnO@SE dosage, and contact time. Adsorbent weight (M in g) and volume (V in L) and initial and final metal ion concentration (Cr and Pb in mg/L) were taken into account, and the mass balance method for metal ion adsorption was calculated with Eq. (1).

$$Y_t = \frac{V}{M} (C_i - C_e) \quad (1)$$

The effectiveness of a green-synthesized MnO@SE adsorbent in absorbing heavy metal ions from wastewater was investigated, focusing on sustainable water management through mass balance determination, through Eq. (2).

$$\text{Absorption}\% = \frac{C_i - C_e}{C_i} \times 100 \quad (2)$$

2.5 Adsorption Kinetics

Kinetic experiments were conducted to examine the metal ion adsorption pattern by the MnO@SE adsorbent [18]. Maximum adsorption conditions were derived from the batch test, whereupon curves were plotted in determining linearity in the relationship between adsorbent-adsorbate and the physical and chemical phenomena of adsorption. Solid-liquid phase interaction, retention rate correlation, and maximum adsorption conditions were explored using isotherm and linear models. Pollutant adsorption through solid-liquid interactions is implied by pseudo-first-order theory, which indicates that the solute transformation rate is proportional to the adsorbate concentration (Eq. (3)).

$$q_e = q_e (1 - e^{-kt}) \quad (3)$$

The pseudo-second-order model suggests that adsorption of impurities onto the MnO@SE adsorbent is proportional to the quantity of active sites on the adsorbent, and the relevant equation is presented in Eq. (4).

$$\frac{dq}{dt} = k(q_e - q)^2 \quad (4)$$

The study further examined the efficiency of MnO@SE adsorption with gas molecules during the first stage, finding that despite a decrease in desorption rate, metal ions increased exponentially, as computed in Eq. (5).

$$q_t = 1 + \frac{1}{b} \ln(1 + abt) \quad (5)$$

The Boyd kinetic model was used in establishing the step that regulates the rate of the heavy metal biosorption by the biosorbent. The Boyd equation is as presented in Eq. (6), while Di value was calculated from Eq. (7), following the determination of B values from Boyd plots.

$$B = 1 + \frac{\Pi^2 Di}{r^2} \quad (6)$$

$$B_t = -0.4977 - \ln(1 - F) \quad (7)$$

Intraparticle diffusion (IPD) is feasible only at high solute concentrations in batch absorbers. The adsorption process is proven to be caused by intraparticle diffusion if the q_t vs. $t^{1/2}$ plot is from the origin. If a multilinear plot is present, it indicates the involvement of more than one step in the adsorption process. The mathematical representation of IPD studies is illustrated by Eq. (8).

$$q_t = k_p \tau^2 + C \quad (8)$$

2.6 Isothermal Studies

Isothermal studies are crucial for determining the adsorption capacity of green-synthesized MnO NPs under regulated parameters, like temperature, adsorbent dose, and metal ion concentrations. These studies describe the relationship between the maximum adsorption capacity (q_{max} , mg/g), which represents the maximum amount of metal ions adsorbed per unit mass of adsorbent at monolayer coverage, and the equilibrium concentration (C_e) of metal ions in solution. The Langmuir and Freundlich isotherms are widely used to study the interaction between MnO NPs and heavy metals, focusing on the solidliquid phase behaviour and chemical interactions, respectively. It is represented in Eq. (9).

$$q_e = \frac{q_{max} K_L C_e}{1 + K_L C_e} \quad (9)$$

where, q_e (mg/g) is the adsorption capacity at equilibrium, q_{max} (mg/g) is the maximum monolayer adsorption capacity, K_L (L/mg) is the Langmuir constant related to adsorption affinity, and C_e (mg/L) is the equilibrium concentration of metal ions in solution.

The Freundlich isotherm model, however, favours heterogeneous adsorption and indicates the multilayer mechanism. This can be described by Eq. (10).

$$\log q_e = \log K_F + \frac{1}{n} \log C_e \quad (10)$$

where, K_F represents the Freundlich capacity constant, indicating the level of equilibrium concentration, and $1/n$ denotes the adsorption intensity in a heterogeneous system.

Beyond Langmuir and Freundlich, other isothermal models, such as Sips, Toth, and Fritz-Schlunder are utilized to analyze pollutant adsorption characteristics. The Sips model integrates aspects of both the Langmuir and Freundlich isotherms, predicting heterogeneous site presence in adsorption behaviour and is represented in Eq. (11).

$$\frac{1}{q_e} = 1 + \frac{1}{q_{max} K_s} \left(\frac{1}{C_e} \right)^{1/n} + \frac{1}{q_{max}} \quad (11)$$

The Toth isotherm model addresses discrepancies between experimental data and the Langmuir model, explaining the adsorption process across a range of metal ion concentrations, as expressed in Eq. (12).

$$\ln \frac{q_e}{q_m - q_e} = n \ln K_L + n \ln C_e \quad (12)$$

where, n is the exponent in the Toth model, K_L is the model constant, and C_e represents the concentration of the adsorbate at equilibrium.

Finally, the four-parameter model, also referred to as the Fritz-Schlunder isotherm model, is employed to interpret a wide range of experimental data and optimize the adsorption process. Eq. (13) provides the expression for its equation.

$$q_e = \frac{qm_{FS} K_{FS} C_e}{1 + qm_{FS} C_e N_{FS}} \quad (13)$$

The parameters of this model can be ascertained through nonlinear regression analysis, offering a comprehensive comprehension of the adsorption behaviour. In this case, N_{FS} stands for the exponent model, K_{FS} for the equilibrium constant, and qm_{FS} for the maximum adsorption capacity.

3 Results and Discussion

3.1 Characteristics of Green-Synthesized Nanoparticles

The chemical composition and structure of the MnO@SE NPs were determined by various analytical techniques. FTIR identified functional groups, like O–H, C=O, N–H, and C–O, with a characteristic Mn–O peak at approximately 580 cm^{-1} , confirming the capping of NPs by bioactive phytomolecules from the plant extract (Figure 1a). Correspondingly, in the XRD pattern, these distinct diffraction peaks corresponding to the (101 and 102) crystallographic planes matched very well with JCPDS No. 78-0424 and hence confirmed the crystalline nature with high purity of the prepared NPs (Figure 1b). As depicted in Figure 1c, EDX analysis of the synthesized MnO@SE NPs confirmed MnO NPs supported by SE, with well-defined peaks for manganese, oxygen, and carbon, which indicated that plant biomolecules had been adsorbed onto the surface of the NPs without impurities. The SEM images represent the surface morphology of MnO@SE NPs, showing a predominantly spherical morphology, with rather uniform distribution and an approximate average size of $80 \pm 0.5\text{ nm}$ (Figure 1d).

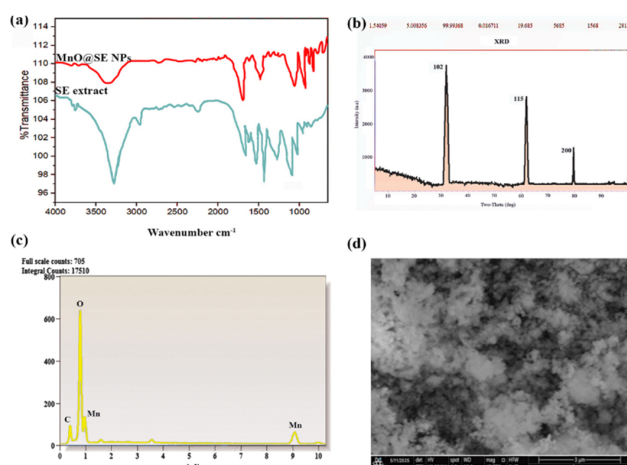


Figure 1. Characteristics of the green-synthesized manganese oxide (MnO) nanoparticles (NPs): (a) Fourier transform infrared spectroscopy (FTIR) spectra; (b) X-ray diffraction (XRD) spectra; (c) energy-dispersive X-ray spectroscopy (EDX) analysis; (d) scanning electron microscopy (SEM) micrographs

3.2 Studies on Adsorption Isotherms

Studies on the metal ion adsorption pattern by the MnO@SE adsorbent were conducted, and the results are presented in Table 1 and Figure 2.

(a) Langmuir isotherm: A Langmuir isotherm study was conducted to analyze the adsorption of metal ions but showed that adsorption did not follow the concept of monolayer [19]. Regression coefficients (R^2) for Langmuir plots were found to be below 0.95, demonstrating their irrelevance in the system.

(b) Freundlich isotherm: In contrast, Freundlich isotherm analysis was consistent with the data, with R^2 values greater than 0.95. This was a confirmation of multilayer adsorption that occurred through a chemical mechanism, depicting a heterogeneous and dynamic adsorption process [20].

(c) Sips isotherm: The Sips model also fitted well with the adsorption data and depicted the homogeneous and heterogeneous nature of the adsorption activity [21].

(d) Toth isotherm: On the other hand, the Toth isotherm, which was created for heterogeneous surfaces, yielded lower regression values, indicating that it was less adapted to this adsorption scenario compared to the other models [22].

(e) Fritz-Schlunder isotherm: Lastly, the Fritz-Schlunder isotherm exhibited high regression values and well described the adsorption under all conditions, confirming the efficiency of the biosorption process. Overall, the findings demonstrated that metal ion adsorption onto MnO@SE was governed by a multistep of parameters, such as concentration, contact time, and surface nature, and provided valuable input for the optimization of wastewater treatment strategies [23].

Besides R^2 , root mean square error (RMSE) and chi-square (χ^2) values were also assessed to validate the reliability of the isotherm models. The Freundlich and Sips models, with the lowest RMSE (0.041–0.058) and χ^2 (0.003–0.005), showed an appropriate fit for Cr(VI) and Pb(II), further confirming their best fit to experimental data. On the other hand, the Langmuir and Toth models presented higher RMSE and χ^2 errors, as was expected from the lower values of R^2 (<0.90). These results further confirm that multilayer heterogeneous adsorption behaviour best describes the adsorption process.

Table 1. Isothermal constants of adsorption for metal ions

Model	Parameter	Calculated Values		RMSE	χ^2
		Cr(VI)	Pb(II)		
Langmuir	q_{max}	12.453	10.011	0.094	0.012
	K_L	0.502	0.623		
	R^2	0.891	0.844		
Freundlich	K_f	4.022	2.876	0.041	0.004
	n	3.876	3.991		
	R^2	0.950	0.985		
Sips	K_S	15.944	9.029	0.037	0.003
	β_S	1.579	1.988		
	a_S	0.703	0.459		
Toth	R^2	0.973	0.985	0.116	0.018
	q_{max}	33.482	29.992		
	b_T	0.511	0.322		
Fritz–Schlunder	n_T	0.987	0.865	0.045	0.005
	R^2	0.831	0.827		
	qm_{FS}	0.522	0.342		
Fritz–Schlunder	K_{FS}	60.011	189.237	0.045	0.005
	N_{FS}	1.432	0.702		
	R^2	0.954	0.963		

Note: q_{max} : maximum adsorption capacity (mg/g); K_L : Langmuir adsorption constant (L/mg); K_F : Freundlich adsorption constant; n : adsorption intensity; K_S : Sips isotherm constant; β_S and a_S : Sips model parameters; b_T and n_T : Toth isotherm constants; qm_{FS} : maximum adsorption capacity in Fritz-Schlunder model; K_{FS} : Fritz-Schlunder equilibrium constant; N_{FS} : Fritz-Schlunder exponent; RMSE: root mean square error; χ^2 : chi-square test; R^2 : regression coefficient.

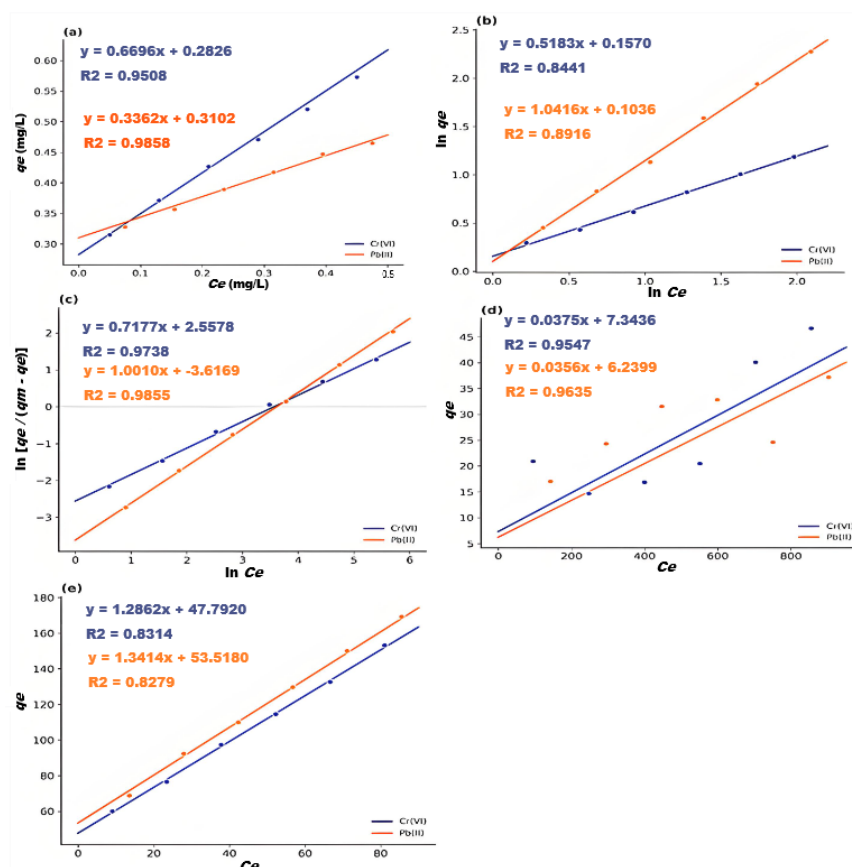


Figure 2. Different models for chromium (Cr) and lead (Pb) adsorption: (a) Freundlich; (b) Langmuir; (c) Sips; (d) Fritz-Schlunder isothermal; (e) Toth linear plots

3.3 Kinetic Studies on Manganese Oxide Nanoparticles from Star Anise Extract

Kinetic investigations were performed to analyze the metal ion adsorption behaviour of Pb(II) and Cr(VI) onto MnO@SE. Varying the initial metal ion concentration in wastewater systematically established the characteristics of the adsorption mechanism: chemical or physical (Figure 3).

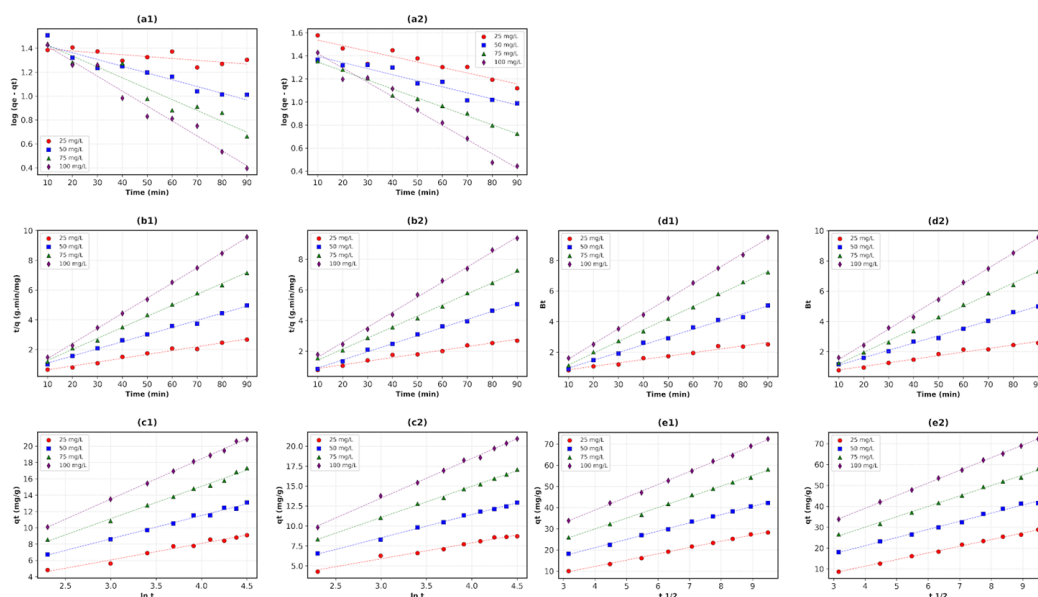


Figure 3. Kinetic models for the adsorption of Cr(VI) and Pb(II) metal ions: (a1) & (a2) Pseudo-first-order; (b1) & (b2) Pseudo-second-order; (c1) & (c2) Elovich model; (d1) & (d2) Boyd model; (e1) & (e2) Intraparticle diffusion (IPD) kinetic plots

(a) Pseudo-first-order (PFO) study: The PFO model was used to explore the metal ions' adsorption on MnO@SE. Both Pb(II) and Cr(VI) plots were found to be linear, as shown in Figure 3a(1) and Figure 3a(2). The regression coefficient of the plots, was below 0.95, indicating that the PFO model is unable to explain how adsorption works. This is an indication that the adsorption process is not physical in origin and is poised to have reached a state of saturation [24].

(b) Pseudo-second-order (PSO) study: As a means to better understand the adsorption mechanism, PSO kinetic studies were undertaken in a wide range of concentrations of ions that varied from 25 to 100 mg/L. The resultant PSO plots from both the metal ions are highlighted in Figure 3b(1) and Figure 3b(2). The regression values for the plots were all higher than 0.95, which indicates high correlation with the adsorption process [25]. The kinetic constants derived from these analyses, confirm the chemical adsorption offered by MnO@SE and support the use of the pseudo-second-order kinetic model.

(c) Elovich kinetic study: The Elovich model was utilized to investigate the kinetics of pollutant adsorption on MnO@SE. The corresponding plots are presented in Figure 3c(1) and Figure 3c(2). Given that the regression values were low ($R^2 < 0.95$) and at odds with the results of the pseudo-second-order investigations, the constants a and b demonstrate that this model is not relevant to the adsorption process. As commendable as the Elovich model is when applied to heterogeneous adsorbents, in this context, it fails to describe MnO@SE adequately.

(d) Boyd kinetic study: Boyd kinetic studies were done to assess the diffusion mechanism involved in the process of adsorption [26]. Plots of Bt versus t are shown in Figure 3d(1) and Figure 3d(2). It would have been implied that intraparticle diffusion is the controlling step if the Boyd plots had been linear and passed through the origin. However, the fact that the Boyd plots did not go through the origin suggests that external or film diffusion plays a major role in controlling the adsorption process with MnO@SE. The low regression coefficients also imply that the Boyd model does not fit.

(e) Intraparticle diffusion (IPD) kinetic study: IPD kinetic analyses were conducted at varying concentrations of metal ions, and the graphical results are shown in Figure 3e(1) and Figure 3e(2). Kinetic constants from the studies are shown in Figure 2. The metal ion uptake vs. time plot would have shown more that the rate-controlling step is intraparticle diffusion had it passed through the origin. Instead, double-slope IPD plots were observed to be the case, representing that there exists more than one phase governing the adsorption mechanism. The process is governed initially by boundary layer effects and secondarily by IPD [27]. In general, the kinetic analysis shows that adsorption of both Cr(VI) and Pb(II) onto MnO@SE is governed by a chemical adsorption process governed by external and intraparticle processes of diffusion.

3.4 Thermodynamic Studies

Thermodynamic experiments were performed to assess the spontaneity of the metal ion adsorption process and whether the process is endothermic or exothermic at constant temperatures. The study was aimed at finding the adsorption behaviour of different temperatures and concentrations of metal ions. The thermodynamic plots of Cr(VI) and Pb(II) at 25, 50, 75, and 100 mg/L are shown in Figure 4a and Figure 4b. Values of ΔH° (change in enthalpy) and ΔS° (change in entropy) were also obtained from these plots and are presented in Table 2. Gibbs free energy change for the adsorption process of the metal ions was also calculated, which provides very useful information regarding the nature of MnO@SE nanomaterial. The spontaneity of the metal ion adsorption is demonstrated by the negative Gibbs free energy values in Table 2. The positive ΔH° values further support the process' endothermic nature. The natural disorder of the liquid and solid phases of the MnO@SE adsorbent is also indicated by the positive ΔS° values (Figure 4). Cumulatively, these results confirm the spontaneity of the metal ion adsorption process to be endothermic, as a testament to the performance ability of the MnO@SE adsorbent [28].

Furthermore, the fairly high positive enthalpy ΔH° shows that Cr(VI) and Pb(II) adsorption on MnO@SE is not of a purely physical nature but involves chemisorption. Such magnitudes typically correspond to stronger interactions, including surface complexation and electron sharing between the metal ions and oxygen-containing functional groups on the MnO@SE surface. Similarly, high positive entropy ΔS° reflects increased randomness at the solid-solution interface; the latter could be regarded as desolvation of metal ions and structural rearrangement of active surface sites accompanying chemisorption. These observations suggest that an endothermic and spontaneous process is attributed to chemical bonding and surface reactivity rather than simple van der Waals interactions.

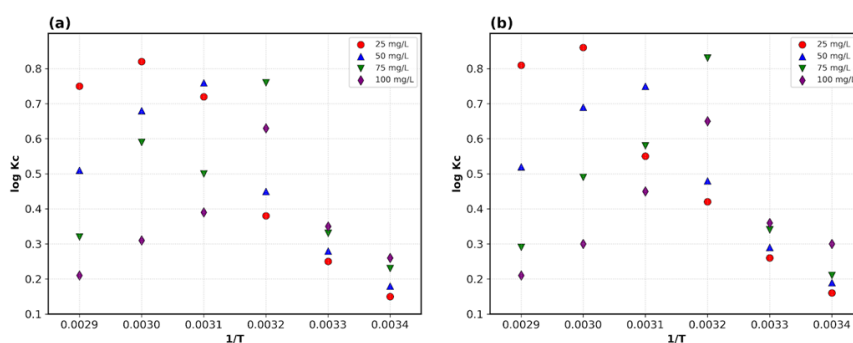


Figure 4. Thermodynamics of manganese oxide nanoparticles synthesized from star anise extract (MnO@SE)-based biosorbent: (a) Cr uptake; (b) Pb uptake

Table 2. Manganese oxide nanoparticles synthesized from star anise extract (MnO@SE) adsorbent thermodynamic constants for metal ion adsorption

Metal Ion Concentration (mg/L)	Enthalpy (ΔH°) kJ/mol	Enthalpy (ΔS°) J/mol·K	Gibbs Energy (ΔG°) kJ/mol			
			20 °C	30 °C	40 °C	50 °C
Cr(VI)						
25	82.356	177.405	-13.563	-11.521	-10.033	-6.871
50	40.332	100.033	-11.678	-9.065	-8.002	-6.002
75	26.004	48.006	-9.007	-8.003	-7.009	-6.001
100	15.098	32.771	-6.006	-6.006	-3.005	-5.001
Pb(II)						
25	72.099	159.876	-11.087	-9.999	-9.0237	-8.221
50	40.001	82.982	-11.336	-10.445	-8.028	-6.887
75	25.099	39.008	-9.0521	-9.088	-6.952	-4.875
100	17.001	20.887	-5.992	-6.087	-4.995	-3.664

3.5 Adsorption Mechanism

Figure 5 schematically depicts how metal ions get adsorbed on the adsorbent (MnO@SE) used in this research. Metal ion adsorption was conducted via IPD as well as via external film diffusion, which in certain conditions could occur simultaneously.

There were three phases of the adsorption process identified within this research. The first phase was regulated by the external film diffusion, where the adsorption and migration of Pb(II) and Cr(VI) ions were controlled by external forces acting on the MnO@SE adsorbent [29]. The second phase was regulated by particle diffusion, where the contaminants diffused into the inner structure of the adsorbent material. The third phase had the added penetration of metal ions through accessible binding sites and enhanced diffusion rates [30].

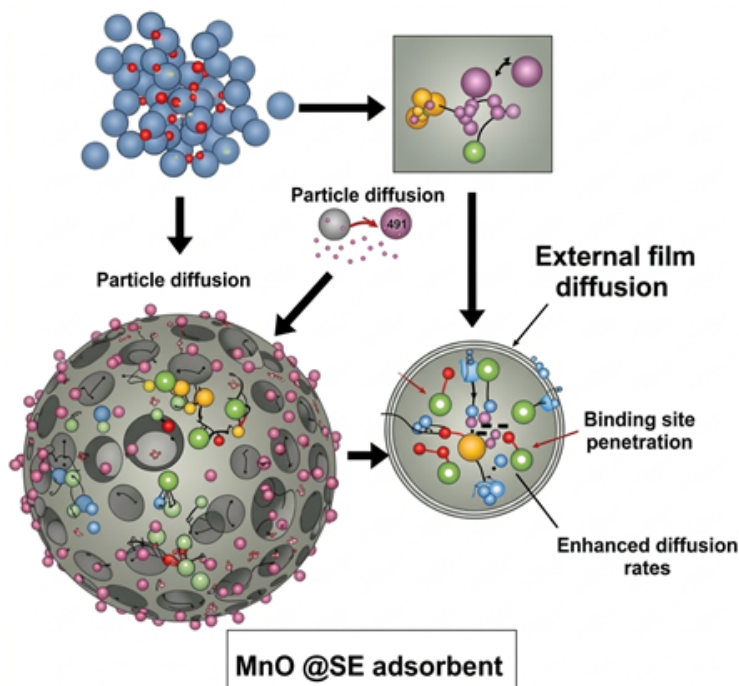


Figure 5. Mechanism of the targeted metal ions' adsorption

The synergistic interactions were responsible for the adsorption of Cr(VI) and Pb(II) onto the MnO@SE adsorbent, as illustrated in Figure 2. Complication of the surface was mainly driven by the carbonate ($-\text{C}=\text{O}$) functional groups that coordinated with the metal ions via their π bonds [31]. This is proven through FTIR analysis where there is a reduction in intensity of the $-\text{CO}_3^{2-}$ peak during adsorption, indicating the involvement of the groups in the uptake process. Besides that, hydrogen bonding with interaction between the oxygen of the $-\text{C}=\text{O}$ group and hydrogen of HCrO_4^- was achieved. Protonation of the adsorbent surface brought upon by the acidic condition triggered a positive charge that offered electrostatic attraction between the MnO@SE adsorbent and Cr(VI) [32]. The Cr(VI) and Pb(II) ions were effectively removed through the overall adsorption process, which was identified as ion pair diffusion (IPD). When the pH was adjusted to highly alkaline conditions, a yellow precipitate of CaCr_2O_7 was observed forming on the adsorbent surface [33].

The high efficiency of Cr(VI) removal by MnO@SE may be due not only to the synergetic interaction of the MnO surface with the phenolic and carbonyl groups of SE but also because of its high active surface area. Such organic functional groups improve the surface reactivity due to electron transfer and complexation with Cr(VI) species, which favours its reduction to Cr(III) and thus subsequent adsorption. Secondly, as deduced from FTIR and kinetic studies, the heterogeneous surface morphology of MnO@SE allows different adsorption sites to enhance the binding affinity and hence overall removal compared with Pb(II) [30].

3.6 Studies on Reusability and Regeneration

Figure 6a indicates that while the metal ions were recovered at high quantities in the initial cycle of regeneration, the recovery rate reduced in the subsequent cycles. The study achieved percentage recoveries of approximately 75.22% for Cr(VI) and 66.44% for Pb(II). The desorption rate of the metal ions using hydrochloric acid is presented in Figure 6b. The operation started with a high recovery of metal ions that was improved at higher concentrations of HCl. At a hydrochloric acid concentration of 0.3 N, the desorption rate peaked. Any further increment in the concentration of the acid caused the desorption rate to decrease, an indication that equilibrium had been reached. The desorption capacity of the spent MnO@SE adsorbent was positively correlated with the rate of desorption, decreasing notably with concentration. To assess the effectiveness of the MnO@SE adsorbent, multiple regeneration cycles were performed.

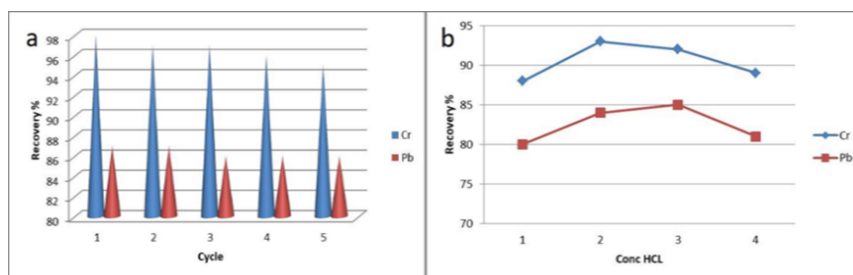


Figure 6. Recycling studies of (a) metal ion adsorption and (b) desorption

Table 3 shows that MnO@SE exhibited better performances in comparison with other MnO-based biosorbents, especially in the case of Cr(VI) (96.34%) and Pb(II) (87.01%). An explanation for this could be that the interaction between MnO NPs and phenolic compounds from SE increased the functionality and surface reactivity towards metal ions. In addition, high values were obtained at fairly low doses of MnO@SE (0.8 g/L) and a rather short contact time (50 min), suggesting fast kinetics of adsorption and effective mass transport. Although experimental conditions varied between studies, it is considered that the green-synthesized MnO@SE nanomaterial represents a competitive, sustainable alternative for heavy metal removal compared to other MnO-based sorbents.

Table 3. Comparison of experimental results with other research works using manganese oxide nanoparticles (MnO NPs) as a biosorbent

Metal	Eff. (%)	Optimal pH	Optimal Dose (g/L)	Initial Metal Ion Conc. (mg/L)	Contact Time (min)	Isotherm Fit	Kinetic Fit	Reference
Ar	70	6.0	0.5	10	80	Langmuir	–	[28]
Cu	60	5.0	0.2	40	40	Langmuir	PSO	[12]
Zn	84	7.0	0.6	9	60	Freundlich	PSO	[17]
Pb and Mn	77 and 90	4.0	1.02	20	200	Langmuir, Freundlich, and Temkin	–	[23]
Pb	82	6.0	0.4	100	40	Freundlich	–	[8]
Pb and Cd	77	4.0	0.6	60	190	Temkin and D–R	PSO	[10]
Cr, Pb and Zn	92.01, 79.11, and 80, respectively	5.0	0.9	80	70	Langmuir	ELO	[9]
Cr and Pb	96.34% for Cr(VI) and 87.01% for Pb(II)	2.0	0.8	25	50	Freundlich	PSO	In this study

4 Conclusions

The effectiveness of MnO@SE NPs in removing Pb and Cr metal ions from wastewater was evaluated through batch adsorption experiments. Removal efficiencies of 96.34% of Cr(VI) and 87.01% of Pb(II) at pH 2.0, a biosorbent dosage of 0.8 g/L, an initial ion concentration of 25 mg/L, and a contact time of 50 minutes were discovered through experiments carried out under ideal conditions. A multilayer adsorption technique was suggested via isothermal investigation that involved heterogeneity of the biosorbent, as observed from the findings of the kinetic studies that indicated an adsorption process governed by diffusion. Thermodynamic studies also corroborated the endothermic and spontaneous characteristics of the adsorption mechanism. Desorption studies indicated that the metal ions adsorbed were quantitatively desorbed with the addition of 0.3 N HCl, indicating biosorbent regenerability. Utilization of MnO@SE NPs derived from a renewable resource is an economic and environmentally friendly approach for the remediation of wastewater heavy metal pollution.

In practical terms, MnO@SE shows tremendous potential for real applications in wastewater treatment systems due to its high adsorption efficiency, relatively low synthesis cost, and the use of renewable plant extract. This gives MnO@SE great feasibility as an eco-friendly alternative compared to commercial adsorbents. Besides that, the material also showed good reusability, maintaining above 80% of the original Cr(VI) and Pb(II) removal efficiency after four cycles of reuse, hence signalling its suitability for repeated industrial use with minimal loss in performance. Such a study demonstrates scaling up MnO@SE for large-scale water remediation, especially in resource-constrained scenarios, a contribution toward developing sustainable, low-cost technologies for water treatment.

Author Contributions

Conceptualization, Q.A.-N.A.K.A.-I. and G.H.; methodology, Q.A.-N.A.K.A.-I. and A.M.A.; software, G.H.; validation, Q.A.-N.A.K.A.-I., G.H. and A.M.A.; formal analysis, Q.A.-N.A.K.A.-I.; investigation, Q.A.-N.A.K.A.-I. and A.M.A.; resources, G.H.; data duration, A.M.A.; writing—original draft preparation, Q.A.-N.A.K.A.-I.;

writing—review and editing, Q.A.-N.A.K.A.-I. and G.H.; visualization, A.M.A.; supervision, Q.A.-N.A.K.A.-I.; project administration, Q.A.-N.A.K.A.-I. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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